Grid Topology Optimization for Secure Operations of Electrical Energy Hubs

EERA Joint Programme on Smart Grids – PhD Candidate Award

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This work has received support from the Belgian Energy Transition Fund, FOD Economy, project DIRECTIONS

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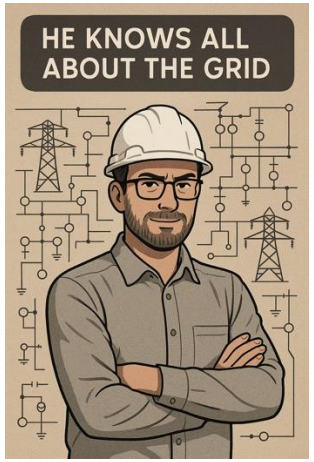
It is time to revolutionize the grid!

Past:

- Conventional generation
- ‘Static’ grid
- Expert-based knowledge of the topological actions in the power system.



[11] GET SET, “Transmission Topology Optimization: State-of-the-Art White Paper”, 2025 . [Online.] Available: https://www.epri.com/research/products/000000003002031445

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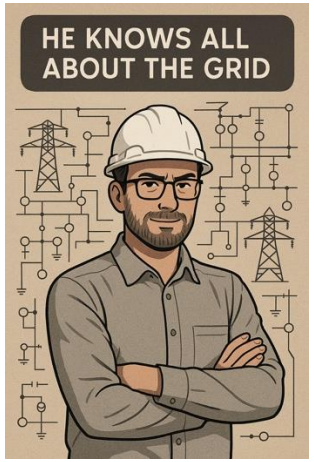
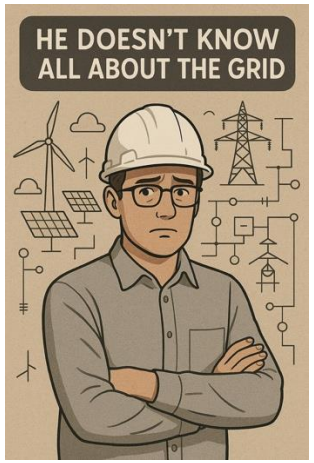
- The fluctuating nature of RES causes variations in the most recurrent power flows in the grid
- ‘Grid enhancing technologies [11]’ (US) or ‘Remedial Action Optimization’ (EU) are being proposed

(!) The grid was not built for variable power flows and distributed energy resources





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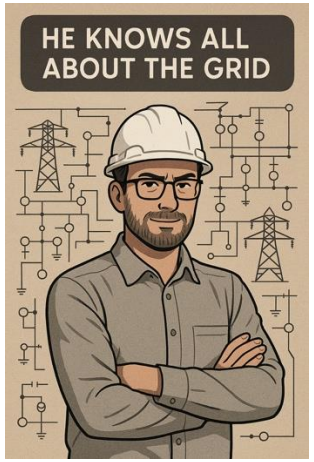

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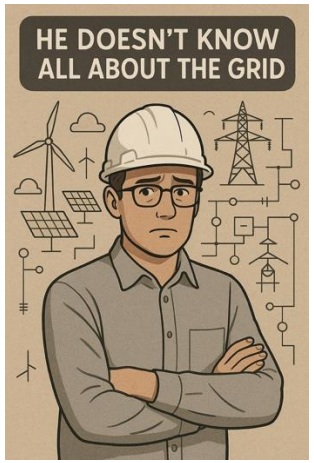
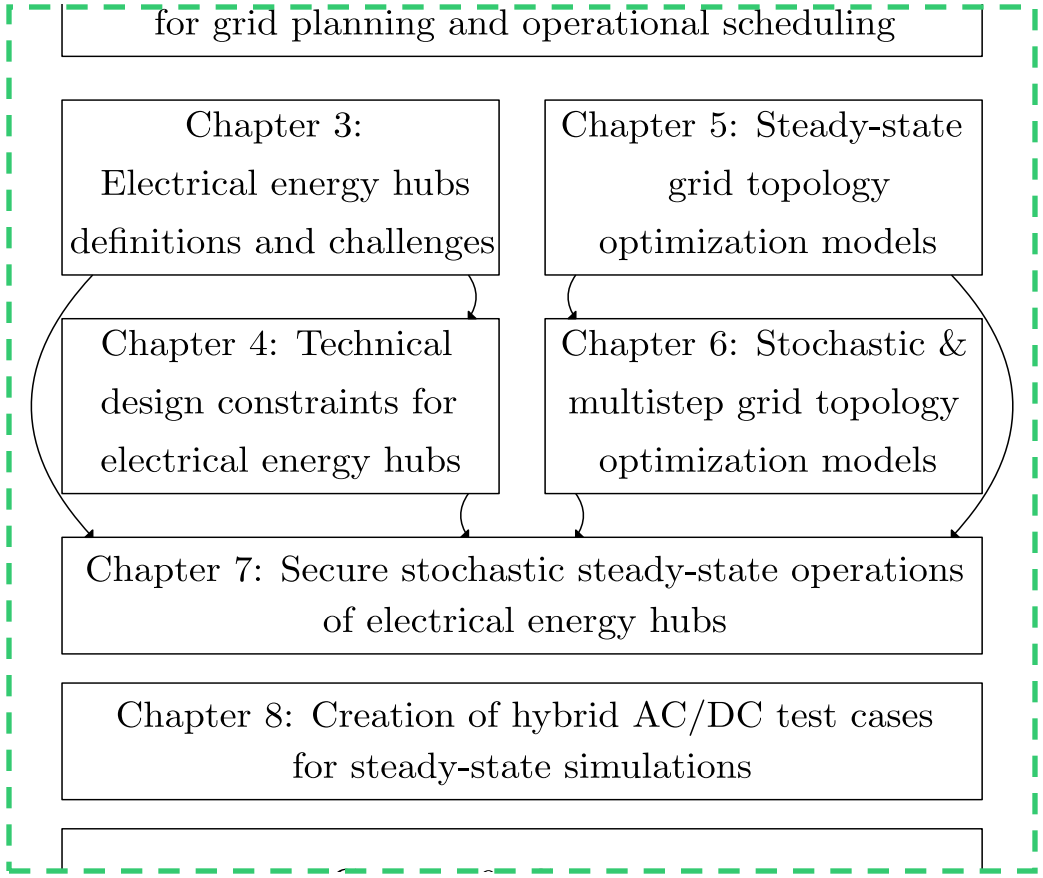


Future:

- Dynamic grid topology optimization to minimize RES generation curtailment and load shedding (plus battery storage and load shifting)
- Coordinated operations of hybrid AC/DC grid to transmit power over long distances
- More efficient grid utilization with *new* technologies such as dynamic line rating, etc.

[11] GET SET, “Transmission Topology Optimization: State-of-the-Art White Paper”, 2025 . [Online.] Available: https://www.epri.com/research/products/000000003002031445

PhD thesis - content



Q1 What is an electrical energy hub? What are the functionalities and technical challenges related to such a concept?

Q2 Can the grid topology be optimized by using topological actions to reduce the total generation costs?

Q3 Can the grid topology be optimized hourly or daily while accounting for renewables' uncertainty?

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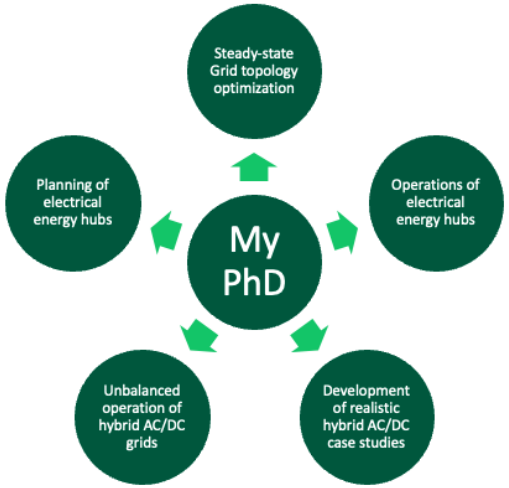

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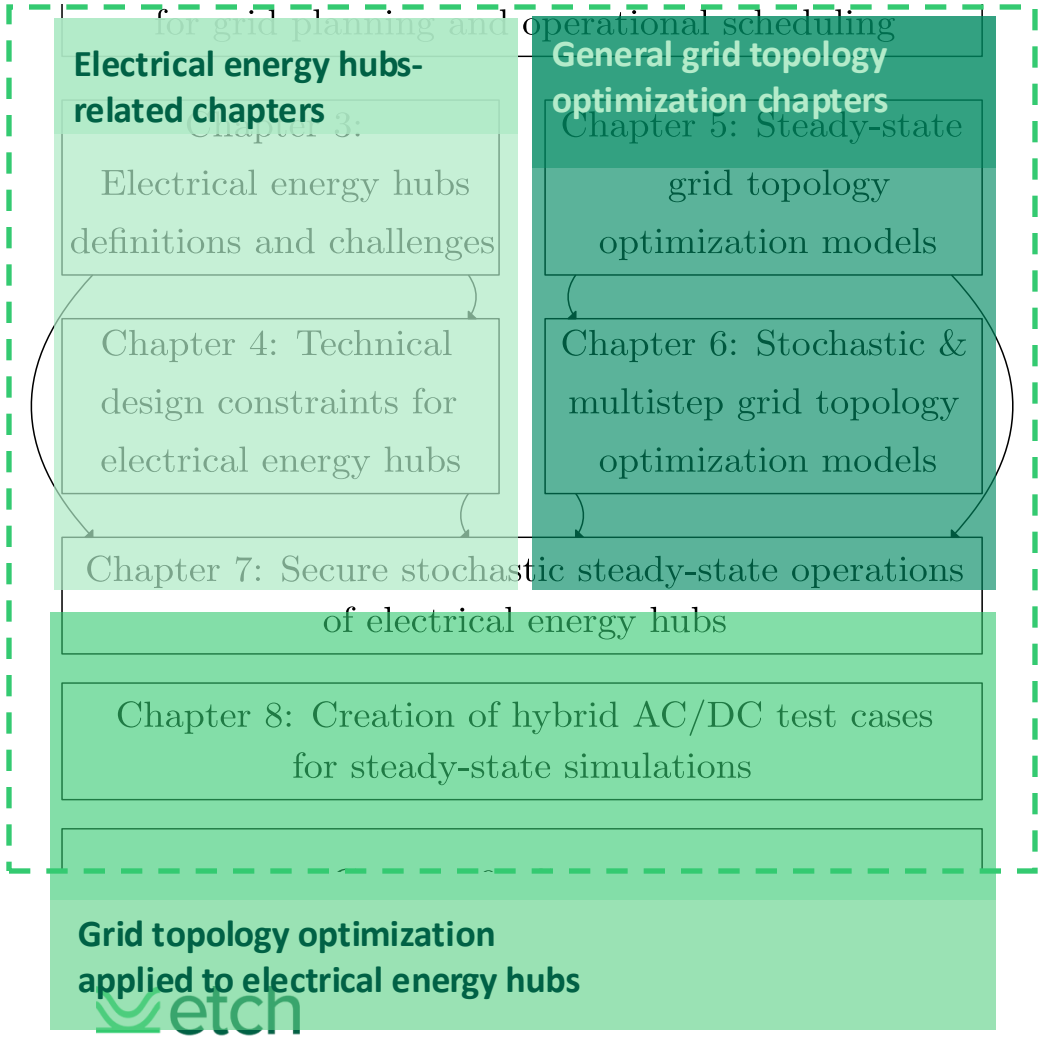

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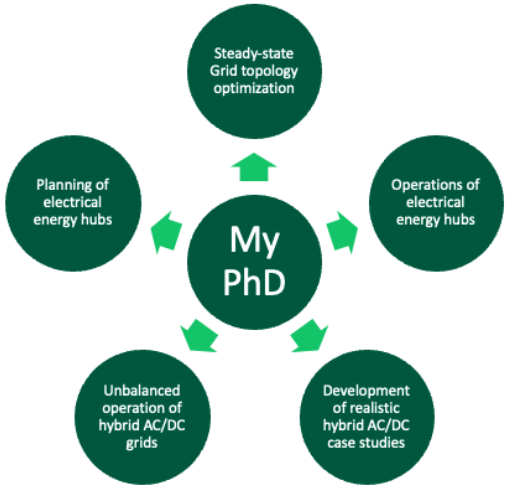
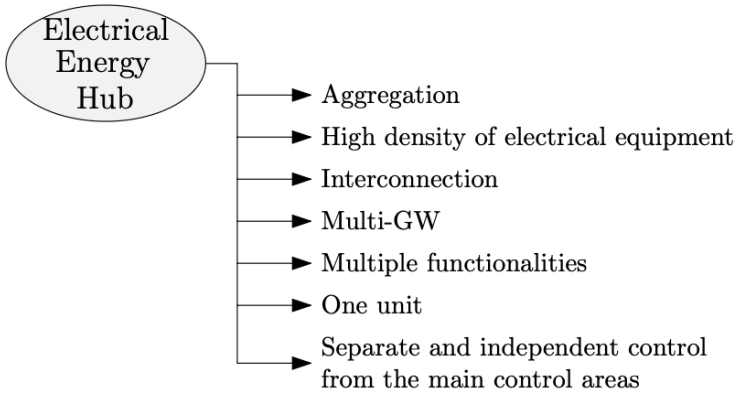
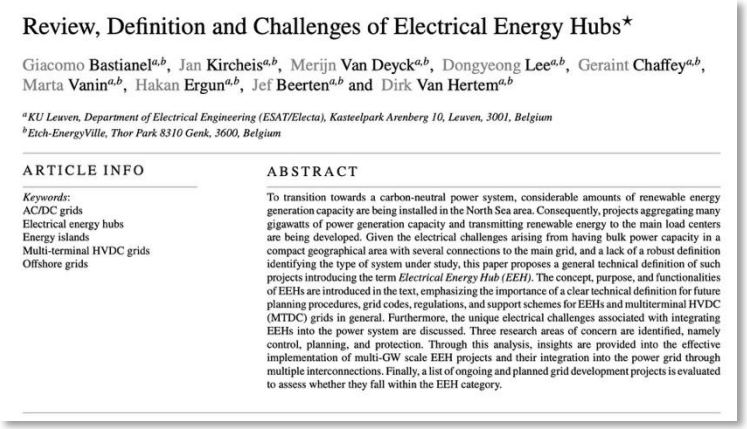


Results for Electrical energy hubs-related chapters

Q1 What is an electrical energy hub? What are the functionalities and technical challenges related to such a concept?

*An **Electrical Energy Hub (EEH)** is a multifunctional node in a power system, managed separately from the main existing control areas, which aggregates local GW-size electrical generation capacity. It is characterized by a high density of electrical equipment and has multiple interconnections to other control areas.*





Review, Definition and Challenges of Electrical Energy Hubs*

Giacomo Bastianel[a,b], Jan Kircheis[a,b], Merijn Van Deyck[a,b], Dongyeong Lee[a,b], Geraint Chaffey[a,b], Marta Vanin[a,b], Hakan Ergun[a,b], Jef Beerten[a,b] and Dirk Van Hertem[a,b]

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ARTICLE INFO

Keywords:
AC/DC grids
Electrical energy hubs
Energy islands
Multi-terminal HVDC grids
Offshore grids

ABSTRACT

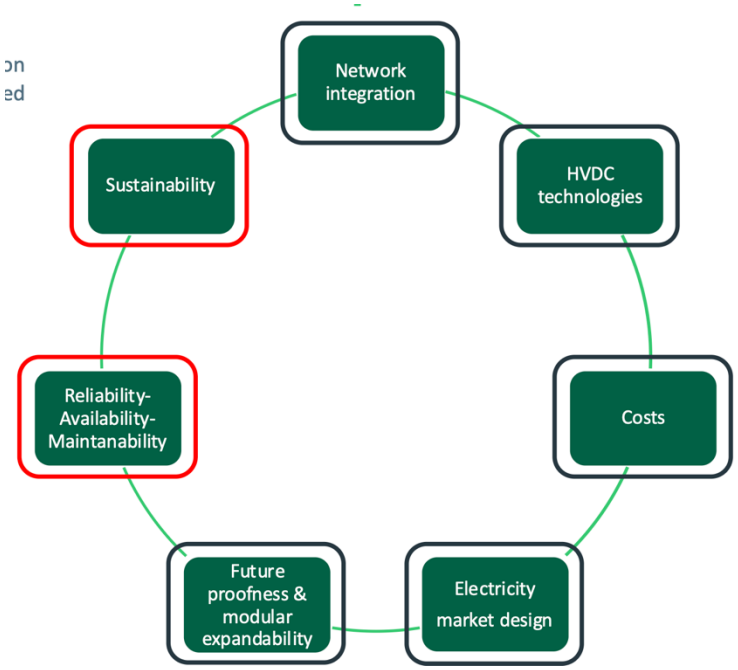
To transition towards a carbon-neutral power system, considerable amounts of renewable energy generation capacity are being installed in the North Sea area. Consequently, projects aggregating many gigawatts of power generation capacity and transmitting renewable energy to the main load centers are being developed. Given the electrical challenges arising from having bulk power capacity in a compact geographical area with several connections to the main grid, and a lack of a robust definition identifying the type of system under study, this paper proposes a general technical definition of such projects introducing the term *Electrical Energy Hub (EEH)*. The concept, purpose, and functionalities of EEHs are introduced in the text, emphasizing the importance of a clear technical definition for future planning procedures, grid codes, regulations, and support schemes for EEHs and multiterminal HVDC (MTDC) grids in general. Furthermore, the unique electrical challenges associated with integrating EEHs into the power system are discussed. Three research areas of concern are identified, namely control, planning, and protection. Through this analysis, insights are provided into the effective implementation of multi-GW scale EEH projects and their integration into the power grid through multiple interconnections. Finally, a list of ongoing and planned grid development projects is evaluated to assess whether they fall within the EEH category.

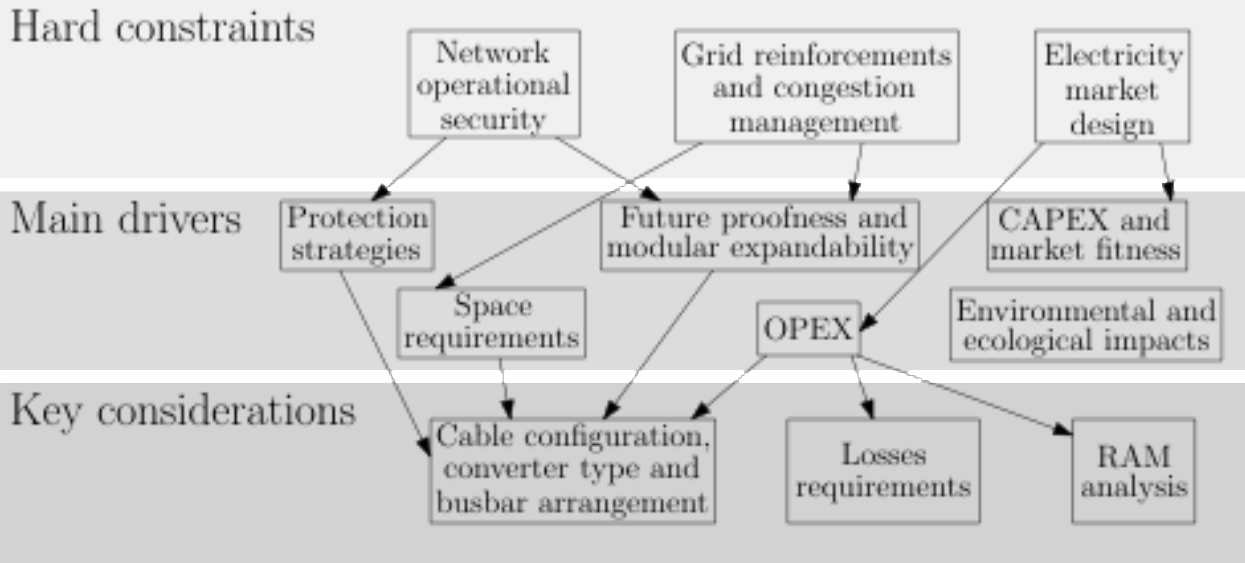
Table 2
Electrical energy hubs (EEHs) are characterized by the following several key functionalities: *aggregation*, *high density of electrical equipment*, *interconnection*, *multi-GW*, *multiple functionalities*, *one unit*, and *separate and independent control from the main control areas*. Several existing and proposed grid expansion projects and their main characteristics are introduced here to assess whether they can be categorized as EEHs based on the proposed functionalities. If the project has all the listed functionalities, it is classified as an EEH. To keep the table concise, the projects' displayed characteristics relate to *aggregation* and *interconnection*, as they consist of the most relevant functionalities for a grid expansion project.

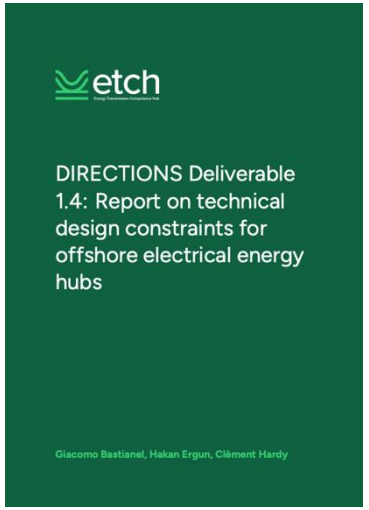
Name	Characteristics	EEH?
Elia's Modular Offshore Grid 1 [6]	2.262 GW offshore wind AC connections to Belgium No interconnections with multiple control areas	✗
Princess Elisabeth Island [12]	3.5 GW offshore wind AC connections to Belgium HVDC interconnections to Belgium, the UK, and a Northern country Located on an artificial island	✓
Danish North Sea energy island [14]	Multi-GW offshore wind HVDC interconnections to Denmark and other European countries Located on an artificial island	✓
Bornholm energy island [13]	3 GW offshore wind HVDC interconnections to Denmark and Germany Located on a natural island	✓
North Sea Wind Power Hub [27]	Hub-and-spoke modular concept Multiple country efforts Interconnections within different hubs	✓
HeideHub [96]	Multi-GW offshore wind Multiple HVDC interconnections Aggregation and interconnection are not in the same node	✗
Shetland MTDC [29]	443 MW onshore wind First MTDC grid in Europe Aggregation and interconnection are not in the same node	✗
StromNetzDC [97]	Multi-GW offshore wind Multiple HVDC interconnections Aggregation and interconnection are not in the same node	✗
Netherton Hub [98]	Multi-GW offshore wind Multiple HVDC interconnections Aggregation and interconnection are not in the same node	✗

Results for Electrical energy hubs-related chapters

Q4 What are the main technical design constraints for TSOs when considering investments in electrical energy hubs?







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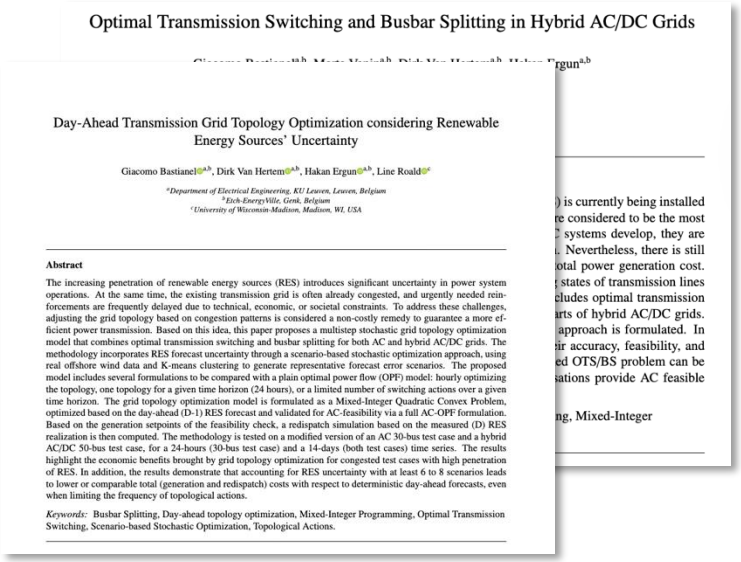

Optimal Transmission Switching and Busbar Splitting in Hybrid AC/DC Grids

Day-Ahead Transmission Grid Topology Optimization considering Renewable Energy Sources' Uncertainty

Giacomo Bastianel[a,b], Dirk Van Hertem[a,b], Hakan Ergun[a,b], Line Roald[c]

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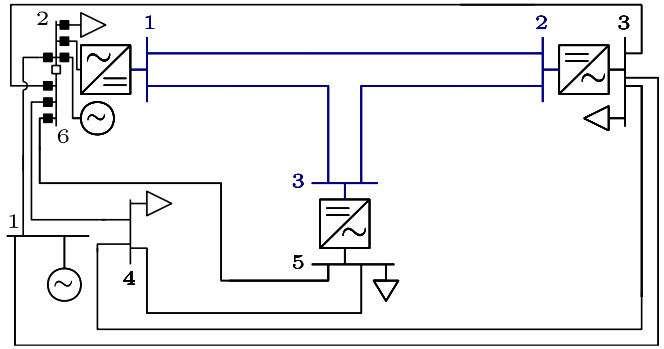
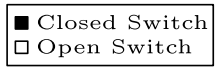
[c]University of Wisconsin-Madison, Madison, WI, USA

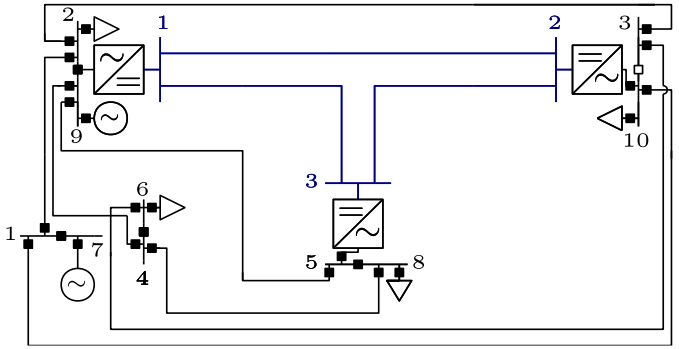
Abstract

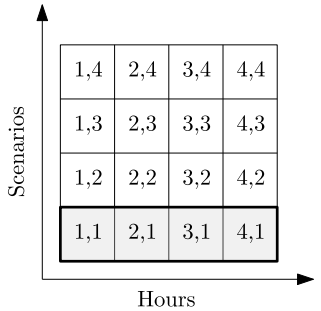
The increasing penetration of renewable energy sources (RES) introduces significant uncertainty in power system operations. At the same time, the existing transmission grid is often already congested, and urgently needed reinforcements are frequently delayed due to technical, economic, or societal constraints. To address these challenges, adjusting the grid topology based on congestion patterns is considered a non-costly remedy to guarantee a more efficient power transmission. Based on this idea, this paper proposes a multistep stochastic grid topology optimization model that combines optimal transmission switching and busbar splitting for both AC and hybrid AC/DC grids. The methodology incorporates RES forecast uncertainty through a scenario-based stochastic optimization approach, using real offshore wind data and K-means clustering to generate representative forecast error scenarios. The proposed model includes several formulations to be compared with a plain optimal power flow (OPF) model: hourly optimizing the topology, one topology for a given time horizon (24 hours), or a limited number of switching actions over a given time horizon. The grid topology optimization model is formulated as a Mixed-Integer Quadratic Convex Problem, optimized based on the day-ahead (D-1) RES forecast and validated for AC-feasibility via a full AC-OPF formulation. Based on the generation setpoints of the feasibility check, a redispatch simulation based on the measured (D) RES realization is then computed. The methodology is tested on a modified version of an AC 30-bus test case and a hybrid AC/DC 50-bus test case, for a 24-hours (30-bus test case) and a 14-days (both test cases) time series. The results highlight the economic benefits brought by grid topology optimization for congested test cases with high penetration of RES. In addition, the results demonstrate that accounting for RES uncertainty with at least 6 to 8 scenarios leads to lower or comparable total (generation and redispatch) costs with respect to deterministic day-ahead forecasts, even when limiting the frequency of topological actions.

Keywords: Busbar Splitting, Day-ahead topology optimization, Mixed-Integer Programming, Optimal Transmission Switching, Scenario-based Stochastic Optimization, Topological Actions.









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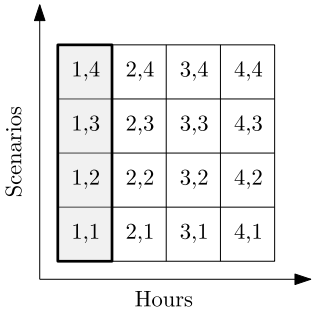
Multistep



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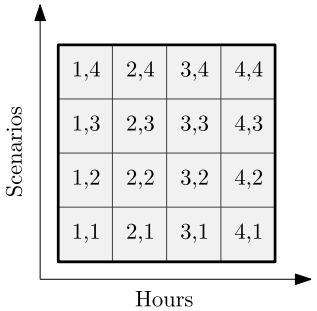
Stochastic



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1,3	2,3	3,3	4,3
1,2	2,2	3,2	4,2
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Multistep & stochastic

Results for general grid topology optimization chapters

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Identifying Best Candidates for Busbar Splitting

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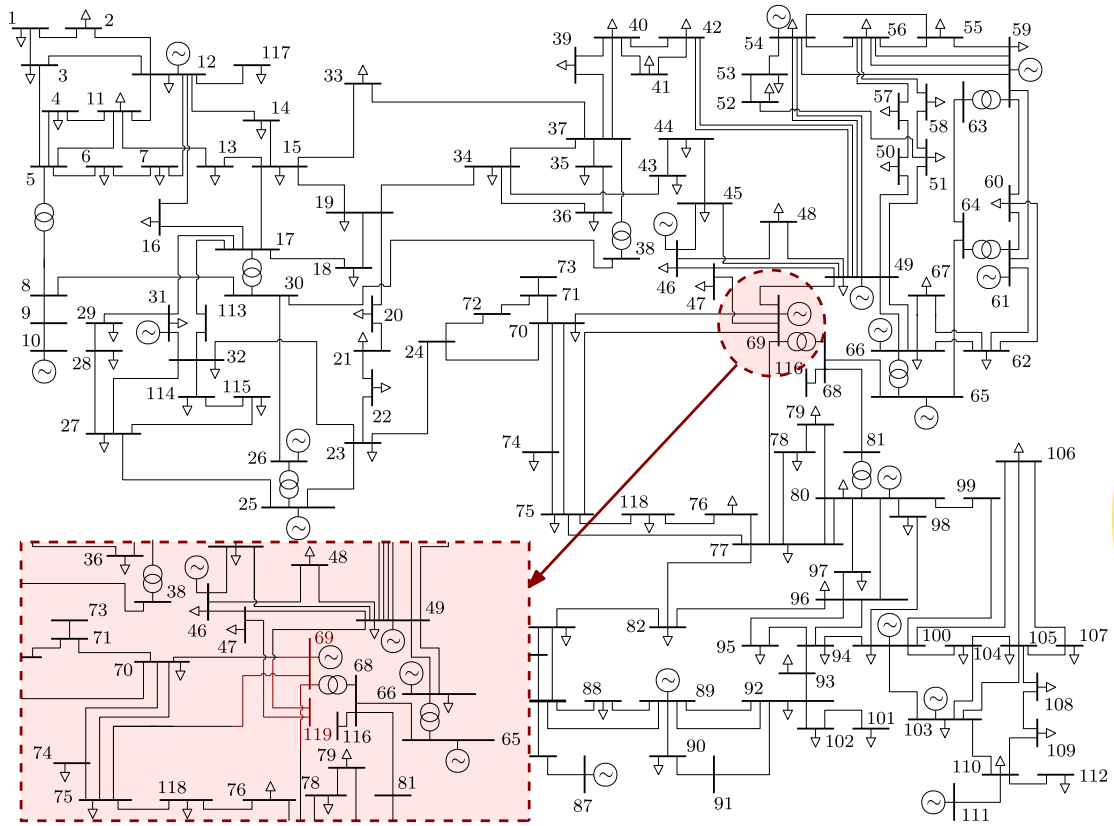
Proposed metrics:

- ΔLMP Generation costs reduction
- # of elements connected to the busbar
- Sum of the absolute differences in LMPs over the branches connected to a busbar (ϕ)
- # of congested branches connected to the busbar
- Voltage angles and magnitudes limits



We developed software to control the power flows in the grid through topological actions

Remedial Action Preventive

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JAO
Joint Allocation Office

Publication Tool

Core CCR

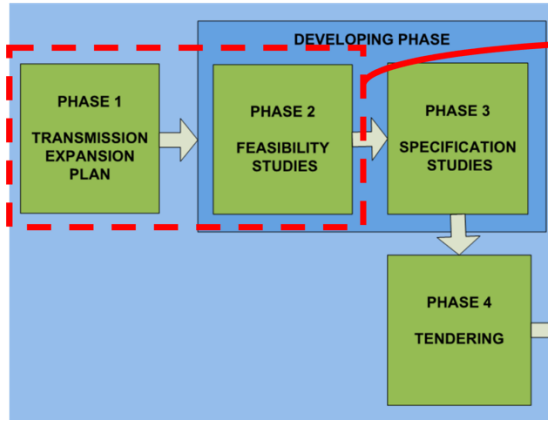
CoreID CCR

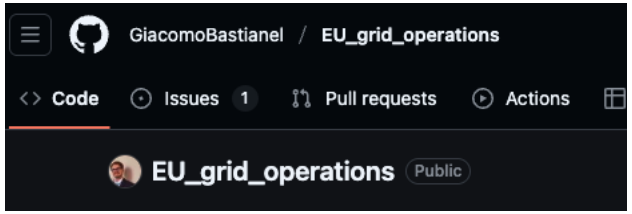
Italy North CCR & IBWT IDA

etch
Energy Transmission Competence Hub

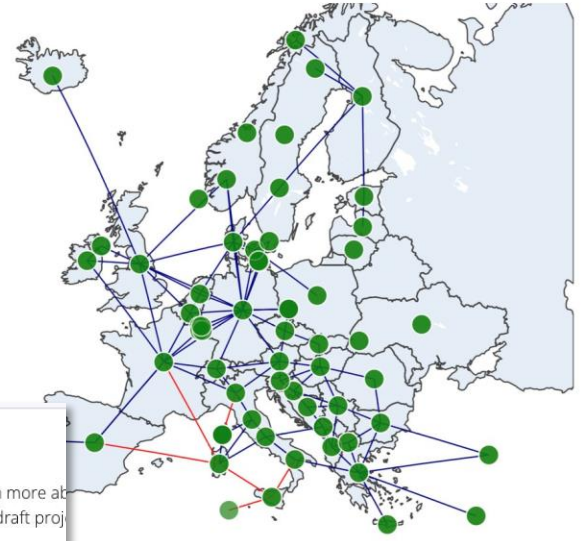
KU LEUVEN

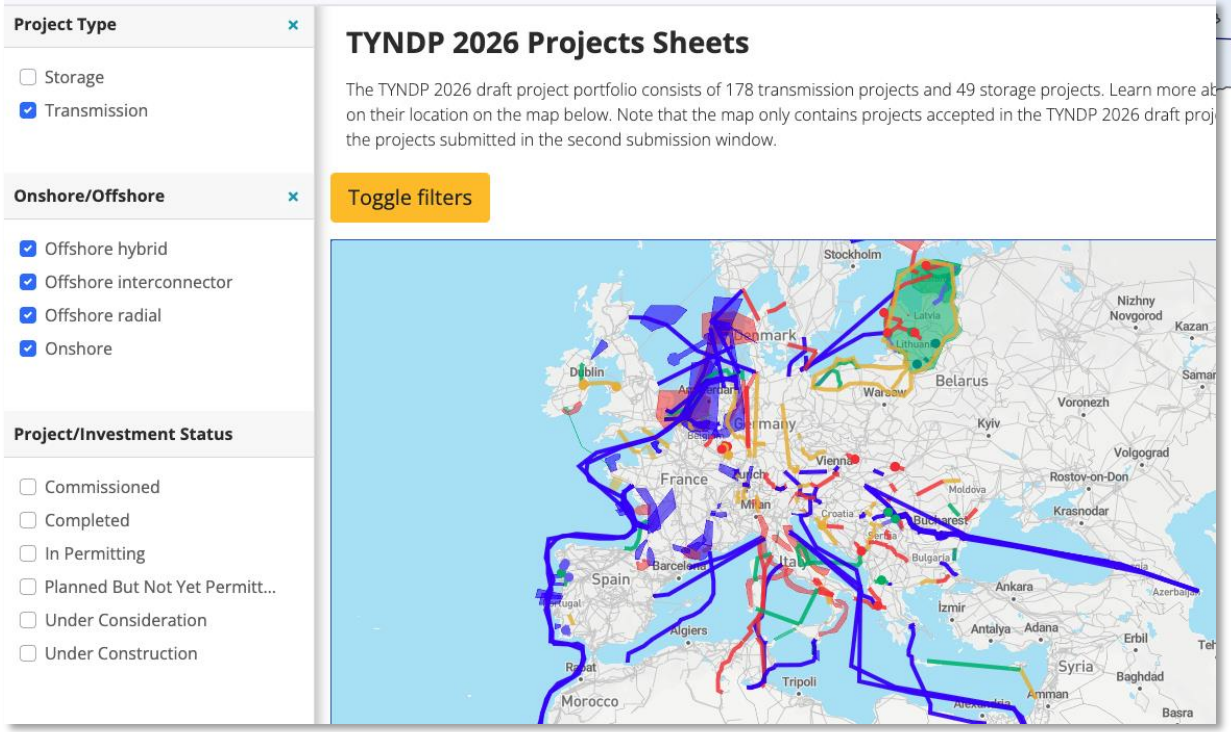
We developed software to find the right investments in grid reinforcements/expansions

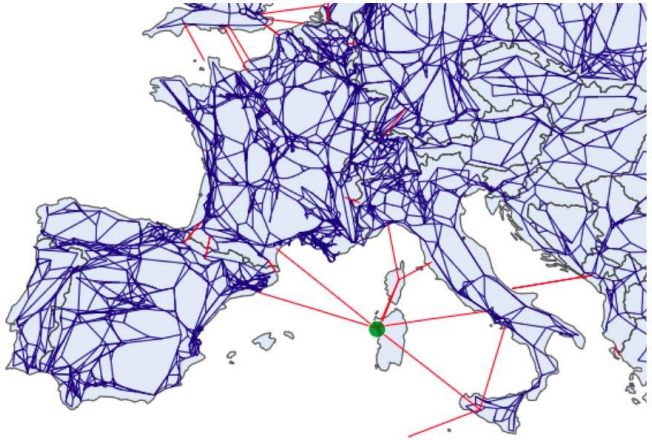




Based on real-life inspired test cases!







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Energy Transmission Competence Hub

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Thank you for your attention!
Any questions?

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Energy Transmission Competence Hub

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